Bhumika Khadka

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Education

East Central University Ada, Oklahoma
Bachelor of Science in Computer Science; Minor: Information Systems. Dean's List; President's Honor Roll (Spring 2026); International Tuition Grant ($10,000/yr, 2024–2026); Member, Alpha Chi National Honor Society. GPA: 3.63. *Graduation Date: Dec 2027*
Relevant Coursework: Data Structures, Database Design, Operating Systems, Cloud Fundamentals, Object-Oriented Programming, Computer Architecture, Web Development

Experience

Market Research Analyst Intern / Data Content Specialist May 2023 – Dec 2023
Global Opportunities Nepal *Kathmandu, Nepal*

- Cleaned and wrangled raw US university data using Python (pandas, numpy), merged institutional and program-level datasets, filtered for STEM-designated programs, imputed missing values, and standardized variable names across multi-year cohorts.
- Analyzed admission trends for international students and automated executive-ready summary reports using openpyxl and python-docx, highlighting top university recommendations.
- Visualized findings using Matplotlib, including bar graphs of top-choice universities and scatter plots mapping tuition costs vs. post-graduation earnings, supporting placement of more than 50 students.
- Researched US study abroad trends and produced TikTok video content and ad templates, growing followers by more than 2,000 and reaching more than 1,000 views.

Projects

Customer Churn Predictor Oct 2026
React, Vite, FastAPI, Python, JavaScript, Git

- Built a full-stack web application with an asynchronous FastAPI RESTful backend and a responsive React (Vite) single-page frontend.
- Trained a Random Forest classifier on tabular customer data, serialized it with joblib, and loaded it into memory at FastAPI startup for low-latency real-time predictions.
- Configured REST endpoints to handle JSON request payloads for ML model inference and set CORS policies for secure cross-origin communication.
- Managed client-side state and async network calls using React hooks, with immediate input validation and dynamic UI feedback.

Bhumika's Travel Journal Oct 2026
Python, FastAPI, SQLite, Leaflet.js, GeoJSON, Pillow, HTML/CSS, JavaScript

- Automated photo metadata ingestion using Python's Pillow library to extract EXIF data (latitude, longitude, timestamps) and reverse-geocode coordinates via OpenStreetMap's Nominatim API.
- Loaded GeoJSON boundary data into a Leaflet.js map to render visited US states with translucent overlays and clickable tooltips.
- Built a timeline sidebar with live text search, state filtering, thumbnail previews, and flyTo animations linked to map markers.
- Set up FastAPI endpoints and SQLite schemas to handle image uploads, return location data, and track visited states.

AI-Powered Grammar Assistant March 2026
Python, OpenAI API, Prompt Engineering

- Built a Python CLI app that corrects grammar and syntax in real time using the OpenAI API, running in a continuous input loop.
- Controlled output by setting a system role with negative constraints, wrapping user input in double-quote delimiters, and setting temperature to 0.2 for consistent results.
- Added input validation and error handling to catch empty inputs and edge cases without crashing.

Skills

Languages: Python, JavaScript, SQL, HTML/CSS, C
Technologies / Frameworks: React, FastAPI, Vite, Leaflet.js, SQLite, Pillow
Developer Tools: Git, GitHub, VS Code, Google Cloud Platform
AI / ML Tools: OpenAI API, Prompt Engineering
Data / Visualization: Matplotlib, Tableau, Excel

Awards & Honors

- Best Beginner Build (Spring 2026): Personal Portfolio Website, Programiz March Challenge
- Wall of Inspiration Feature (Spring 2026): Hatch and Hunt recognized by Programiz for UI/UX design and originality